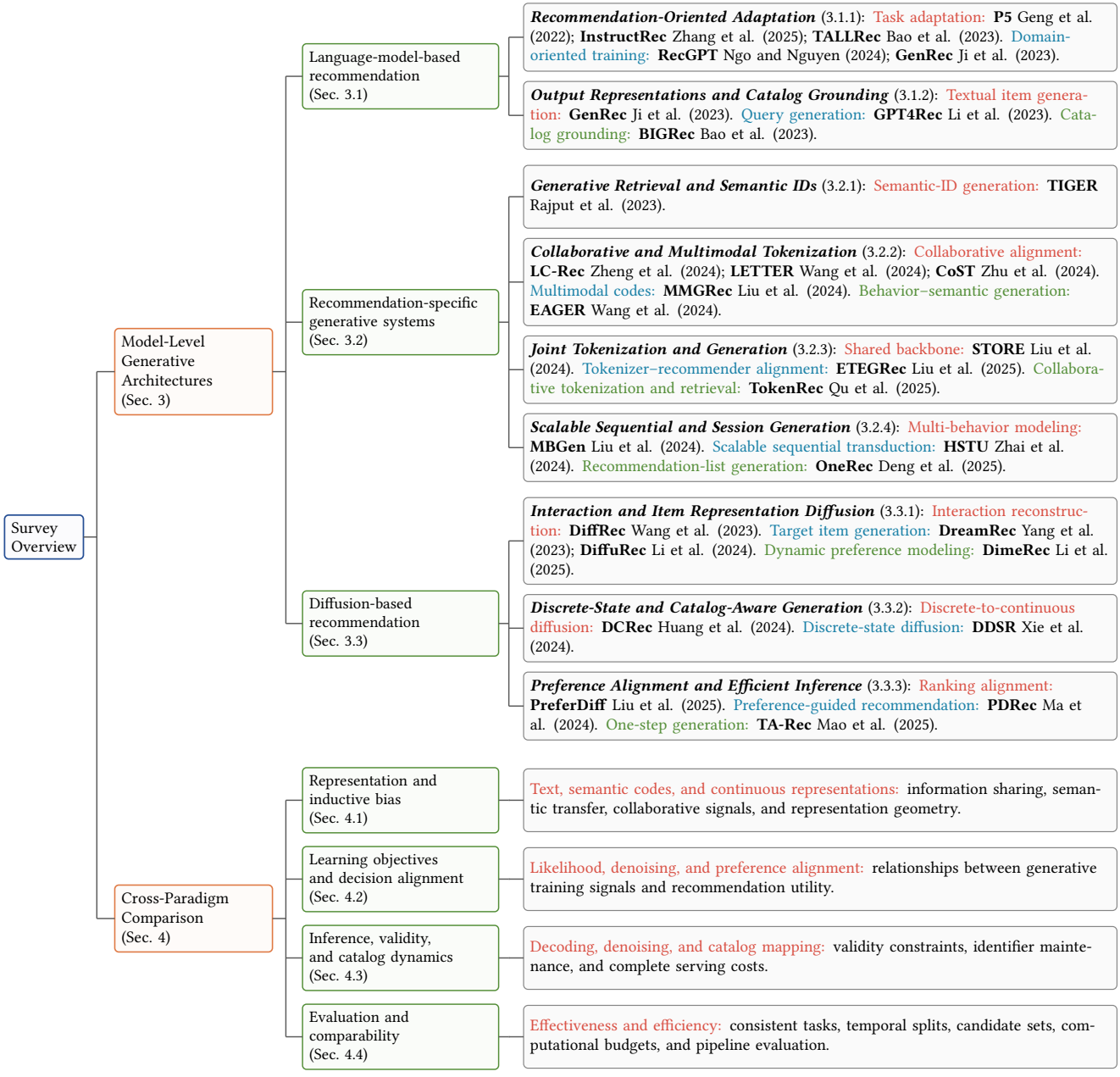

Survey Overview
Model-Level Generative Architectures (Sec. 3)
Language-model-based recommendation (Sec. 3.1)
Recommendation-Oriented Adaptation (3.1.1): Task adaptation: P5 Geng et al. (2022); InstructRec Zhang et al. (2025); TALLRec Bao et al. (2023). Domain-oriented training: RecGPT Ngo and Nguyen (2024); GenRec Ji et al. (2023).
Output Representations and Catalog Grounding (3.1.2): Textual item generation: GenRec Ji et al. (2023). Query generation: GPT4Rec Li et al. (2023). Catalog grounding: BIGRec Bao et al. (2023).
Recommendation-specific generative systems (Sec. 3.2)
Generative Retrieval and Semantic IDs (3.2.1): Semantic-ID generation: TIGER Rajput et al. (2023).
Collaborative and Multimodal Tokenization (3.2.2): Collaborative alignment: LC-Rec Zheng et al. (2024); LETTER Wang et al. (2024); CoST Zhu et al. (2024). Multimodal codes: MMGRec Liu et al. (2024). Behavior–semantic generation: EAGER Wang et al. (2024).
Joint Tokenization and Generation (3.2.3): Shared backbone: STORE Liu et al. (2024). Tokenizer–recommender alignment: ETEGRec Liu et al. (2025). Collaborative tokenization and retrieval: TokenRec Qu et al. (2025).
Scalable Sequential and Session Generation (3.2.4): Multi-behavior modeling: MBGen Liu et al. (2024). Scalable sequential transduction: HSTU Zhai et al. (2024). Recommendation-list generation: OneRec Deng et al. (2025).
Diffusion-based recommendation (Sec. 3.3)
Interaction and Item Representation Diffusion (3.3.1): Interaction reconstruction: DiffRec Wang et al. (2023). Target item generation: DreamRec Yang et al. (2023); DiffuRec Li et al. (2024). Dynamic preference modeling: DimeRec Li et al. (2025).
Discrete-State and Catalog-Aware Generation (3.3.2): Discrete-to-continuous diffusion: DCRec Huang et al. (2024). Discrete-state diffusion: DDSR Xie et al. (2024).
Preference Alignment and Efficient Inference (3.3.3): Ranking alignment: PreferDiff Liu et al. (2025). Preference-guided recommendation: PDRec Ma et al. (2024). One-step generation: TA-Rec Mao et al. (2025).
Cross-Paradigm Comparison (Sec. 4)
Representation and inductive bias (Sec. 4.1)
Text, semantic codes, and continuous representations: information sharing, semantic transfer, collaborative signals, and representation geometry.
Learning objectives and decision alignment (Sec. 4.2)
Likelihood, denoising, and preference alignment: relationships between generative training signals and recommendation utility.
Inference, validity, and catalog dynamics (Sec. 4.3)
Decoding, denoising, and catalog mapping: validity constraints, identifier maintenance, and complete serving costs.
Evaluation and comparability (Sec. 4.4)
Effectiveness and efficiency: consistent tasks, temporal splits, candidate sets, computational budgets, and pipeline evaluation.